MASTER'S COMPREHENSIVE EXAMINATION

Part II | **Applied (90 minutes)** | **December 5, 2009**

Instructions: Answer any **three** out of the four questions. If you answer four, state which questions you want graded. Open book and open notes.

1. The number of persons bitten by snakes in New Jersey each year is a Poisson process.

   A. If 85 People were bitten in 2008 and 55 were bitten in 2009, *assume that there was no change in intensity of snakebites between 2008 and 2009* and give the best point estimate and 95% confidence interval for the expected annual number of snakebites in New Jersey under the above assumptions.

   B. Dr. X believes that the real mean for snakebites dropped between 2008 and 2009 in New Jersey because of a snakebite intervention initiated in 2009. State the null and alternative hypotheses and test with the appropriate Type-1 error of 0.02 under the above assumptions.

2. The current laboratory test for amounts of "Testro" in a student's blood has too much intraperson measurement variation and a new test for Testro is believed to perform better. Eight Graduate Students each are tested twice for Testro with the Standard Test and twice with the New Test with the following results.

Student	TESTRO STANDARD TEST				TESTRO NEW TEST		
	1st Test	2nd Test	**Mean of Both Tests**		1st Test	2nd Test	**Mean of Both Tests**
1	30	36	**33**		28	24	**26**
2	44	44	**44**		41	43	**42**
3	78	70	**74**		73	78	**75.5**
4	55	61	**58**		60	52	**56**
5	62	67	**64.5**		71	67	**69**
6	39	38	**38.5**		40	40	**40**
7	23	33	**28**		30	26	**28**
8	52	46	**49**		48	44	**46**

Assume that all tests on each person are normally distributed and independent and what is observed from each test j on person i, is $Y_{ij} = X_i + \varepsilon_{ij}$ Where

- X_i is the True value for person i
- ε_{ij} is the measurement error of test j on person i which has mean 0 and variance σ_s^2 for the standard test and σ_n^2 for the "New" Test

A. Give the Best Estimates for σ_s^2 and σ_n^2 from the data

B. State any additional Assumptions needed and test

$H_o : \sigma_s^2 = \sigma_n^2$ against $H_a : \sigma_s^2 > \sigma_n^2$ with a Type 1 Error of 0.10

OVER - TEST CONTNUES ON THE OTHER SIDE →→